THE FUNCTOR OF MAPS FROM A PRODUCT WITH A FINITE SIMPLICIAL SET IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-exists-hom-0-from-simplicial-set (source lines 1626-1641); the proof environment begins two lines below it and is its own proof, closing at source line 1675.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; the simplex category and the simplicial objects of C are the author's surrounding development in the same chapter.

Editorial note. The statement displays one eqnarray describing the functor; it is part of the retained source range and is reproduced as written, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Objects chapter, counted once: for a finite simplicial set U the functor of maps from X times U into a fixed simplicial object is representable.

Allowed standard prerequisites (editorial).

- Simplicial object and its category. A simplicial object is a contravariant functor from the simplex category; maps between simplicial objects are natural transformations, and X times U denotes the degreewise product.. It is the setting of the represented functor and of the degreewise analysis in the proof (source lines 1628-1640, 1644-1660).
- Coproducts and countable limits in the base category. The category C has coproducts of any two objects and limits of countable diagrams.. The coproducts let the degreewise conditions be assembled and the countable limits are what guarantees the existence of the representing object (source lines 1628-1630, 1665-1674).
- Finiteness of the simplicial set in each degree. Every U_n is a finite nonempty set.. It converts the degreewise condition into a finite product indexed by the elements of U_n, which is what makes the limit countable (source lines 1630, 1646-1664).

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Lemma 0.1. *Assume the category $\mathcal{C}$ has coproducts of any two objects and countable limits. Let U be a simplicial set, with U_n finite nonempty for all $n \geq 0$. Let V be a simplicial object of $\mathcal{C}$. Then the functor*

$$\begin{aligned} \mathcal{C}^{opp} &\longrightarrow \mathit{Sets} \\ X &\longmapsto \mathrm{Mor}_{Simp(\mathcal{C})}(X \times U, V) \end{aligned}$$

is representable.

Proof. A morphism from $X \times U$ into V is given by a collection of morphisms $f_u : X \to V_n$ with $n \geq 0$ and $u \in U_n$. And such a collection actually defines a morphism if and only if for all $\varphi : [m] \to [n]$ all the diagrams

$$\begin{array}{ccc} X & \xrightarrow{f_u} & V_n \\ {\scriptstyle \mathrm{id}_X}\downarrow & & \downarrow{\scriptstyle V(\varphi)} \\ X & \xrightarrow{f_{U(\varphi)(u)}} & V_m \end{array}$$

commute. Thus it is natural to introduce a category $\mathcal{U}$ and a functor $\mathcal{V} : \mathcal{U}^{opp} \to \mathcal{C}$ as follows:

(1) The set of objects of $\mathcal{U}$ is $\coprod_{n \geq 0} U_n$,
(2) a morphism from $u' \in U_m$ to $u \in U_n$ is a $\varphi : [m] \to [n]$ such that $U(\varphi)(u) = u'$
(3) for $u \in U_n$ we set $\mathcal{V}(u) = V_n$, and
(4) for $\varphi : [m] \to [n]$ such that $U(\varphi)(u) = u'$ we set $\mathcal{V}(\varphi) = V(\varphi) : V_n \to V_m$.

At this point it is clear that our functor is nothing but the functor defining

$$\lim_{\mathcal{U}^{opp}} \mathcal{V}$$

Thus if $\mathcal{C}$ has countable limits then this limit and hence an object representing the functor of the lemma exist. □